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Understanding Analysis Sections Covered: Sections 4.2-4.3

Definitions and Theorems to include in Analysis Journal:

- Definition 4.2.1 (functional limit)
- Definition 4.2.1B (functional limit, topological version)
- Theorem 4.2.3 (functional limit, sequential criterion)
- Definition 4.3.1 (continuity)
- Theorem 4.3.2 (characterizations of continuity)
- Definition of right/left limits (Exercise 4.2.10)
- Definition of infinite limits (Exercise 4.2.9)

Before Class:

Readings: Alcock, Chapter 7

I recommend breaking your reading up as follows (If you try to do it in one sitting you will have a bad time and understand less):

- 7.1-7.3 about functions
- 7.4-7.6 about definition of continuity
- 7.7 about proofs of continuity
- 7.10-7.11 about functional limits, discontinuity, and future directions
- (Optional) 7.8-7.9 about some theorems we will encounter

Analysis Journal Prompts:

- Alcock jumps straight to continuity before defining the limit of a function. Why do you think she does this? What is different in the definition of a functional limit (compare and contrast the definitions on p. 142).
- Compare and contrast all the different definitions of continuity on pages 132 and 133. Which one makes the most sense to you? Which one makes the least sense to you?

Function Definition, Domain and Codomain

$$f : A \rightarrow \mathbb{R}$$

Continuity: Intuition

Continuity ($\mathbb{R} \rightarrow \mathbb{R}$)

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at $c \in \mathbb{R}$ if...

Continuity ($A \rightarrow \mathbb{R}$)

A function $f : A \rightarrow \mathbb{R}$ is continuous at $c \in A$ if...

Continuity (Topological Version)

A function $f : A \rightarrow \mathbb{R}$ is continuous at $c \in A$ if...

Continuity (Sequential Version)

A function $f : A \rightarrow \mathbb{R}$ is continuous at $c \in A$ if...

1. Prove that the topological version and sequential version of the definitions of continuity are equivalent.

Continuity (Functional Limit Version)

A function $f : A \rightarrow \mathbb{R}$ is continuous at $c \in A$ if...

**This only works if c is a _____ of A .

Functional Limit

Let $f : A \rightarrow \mathbb{R}$ and let c be a limit point of A . We say that $\lim_{x \rightarrow c} f(x) = L$ if...

Functional Limit (Topological Version)

Let $f : A \rightarrow \mathbb{R}$ and let c be a limit point of A . We say that $\lim_{x \rightarrow c} f(x) = L$ if...

Functional Limit (Sequential Version)

Let $f : A \rightarrow \mathbb{R}$ and let c be a limit point of A . We say that $\lim_{x \rightarrow c} f(x) = L$ if...

After Class:

Pick & Prove Problems:

★★ Exercise 4.3.5

Show using the definition of continuity (any variation that you wish) that if c is an isolated point of the set $A \subseteq \mathbb{R}$, then the function $f : A \rightarrow \mathbb{R}$ is continuous at c .

★★ Exercise 4.3.8

Decide if the following claims are true or false, providing either a short proof or counterexample to justify each conclusion. Assume throughout that g is defined and continuous on all of $\mathbb{R}$.

- (a) If $g(x) \geq 0$ for all $x < 1$, then $g(1) \geq 0$ as well.
- (b) If $g(r) = 0$ for all $r \in \mathbb{Q}$, then $g(x) = 0$ for all $x \in \mathbb{R}$.
- (c) If $g(x_0) > 0$ for a single point $x_0 \in \mathbb{R}$, then $g(x)$ is in fact strictly positive for uncountably many points.

★★ Exercise 4.3.9

Assume $h : \mathbb{R} \rightarrow \mathbb{R}$ is continuous on its domain and let $K = \{x : h(x) = 0\}$. Show that K is a closed set.

★★ Exercise 4.2.9 (Infinite Limits)

The statement $\lim_{x \rightarrow 0} 1/x^2 = \infty$ certainly makes intuitive sense. To construct a rigorous definition for an infinite limit statement of this form, we replace the (arbitrarily small) $\epsilon > 0$ challenge with an (arbitrarily large) $M > 0$ challenge:

Definition: $\lim_{x \rightarrow c} f(x) = \infty$ means that for all $M > 0$ we can find a $\delta > 0$ such that whenever $0 < |x - c| < \delta$, it follows that $f(x) > M$.

- (a) Show $\lim_{x \rightarrow 0} 1/x^2 = \infty$ using this definition.
- (b) Now, construct a definition for the statement $\lim_{x \rightarrow \infty} f(x) = L$. Give an example of such a limit.
- (c) Lastly, construct a definition for the statement $\lim_{x \rightarrow \infty} f(x) = \infty$. Give an example of such a limit.

★★ Exercise 4.2.10 (Right and Left Limits)

Calculus courses typically refer to the right-hand limit of a function as the limit obtained by “letting x approach a from the right-hand side.”

- (a) Give a proper definition in the style of Definition 4.2.1 for the right-hand and left-hand limit statements:

$$\lim_{x \rightarrow a^+} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow a^-} f(x) = M.$$

- (b) Prove that $\lim_{x \rightarrow a} f(x) = L$ if and only if both the right and left-hand limits equal L .

Understanding Analysis Sections Covered: Sections 4.3, 4.6

Definitions and Theorems to include in Analysis Journal:

- Corollary 4.3.3 (criterion for discontinuity)
- Theorem 4.3.4 (Algebraic Continuity Theorem)
- Theorem 4.3.9 (composition of continuous functions)
- Sets of discontinuity D_f (section 4.6, top of p. 141)
- Types of discontinuities (section 4.6, p. 142)

Before Class:

Readings: None

Analysis Journal Prompts:

- Consider these weird functions introduced by Alcock:

$$1. f(x) = \begin{cases} 2 & \text{if } x \in \mathbb{Z} \\ 1 & \text{if } x \notin \mathbb{Z} \end{cases}$$

$$2. \text{Dirichlet's function: } g(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

$$3. \text{Modified Dirichlet's function: } h(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

Where are these functions *discontinuous*? Focus on whatever definition you like from last class.

Most functions we like are continuous.

Algebraic Limit Theorem for Functional Limits

Let $f : A \rightarrow \mathbb{R}$ and $g : A \rightarrow \mathbb{R}$ and assume $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$ for some limit point $c \in A$.

$$(i) \lim_{x \rightarrow c} kf(x) = \quad (ii) \lim_{x \rightarrow c} [f(x) + g(x)] = \quad (iii) \lim_{x \rightarrow c} [f(x)g(x)] = \quad (iv) \lim_{x \rightarrow c} \left[\frac{f(x)}{g(x)} \right] =$$

Algebraic Continuity Theorem

Let $f : A \rightarrow \mathbb{R}$ and $g : A \rightarrow \mathbb{R}$ be continuous at a point $c \in A$. Then the following functions are also continuous at c :

$$(i) \quad (ii) \quad (iii) \quad (iv)$$

Composition of Continuous Functions

Given $f : A \rightarrow B$ and $g : B \rightarrow \mathbb{R}$, if f is continuous at $c \in A$, and g is continuous at _____, then

All **rational functions** are continuous on their domains.

Rational function:

Idea: If we prove the **identity function** ($f(x) = x$) is continuous, then we can apply the ACT liberally to get any rational function.

1. Prove the identity function is continuous.

Proof.

□

2. Prove x^2 is continuous without using the ACT.

Scratch work:

Proof.

□

3. Does this mean $f(x) = \frac{1}{x}$ is continuous?

Famously Discontinuous Functions

The set of discontinuities of a function f is denoted D_f .

Criterion for Discontinuity

Let $f : A \rightarrow \mathbb{R}$, and let $c \in A$ be a limit point of A . If there exists a sequence $(x_n) \subseteq A$ such that...

then f is not continuous at c .

Example 1: $f(x) = \begin{cases} 2 & \text{if } x \in \mathbb{Z} \\ 1 & \text{if } x \notin \mathbb{Z} \end{cases}$

Example 2: Dirichlet's function: $g(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$

Example 3: Modified Dirichlet's function: $h(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$

Example 4: Step Function: $f(x) = \llbracket x \rrbracket$ (greatest integer function)

Types of Discontinuities

1. removable discontinuity:
2. jump discontinuity:
3. essential discontinuity:

Example 5: Topologist's Sine Function: $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ given by $f(x) = \begin{cases} \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Example 6: $g : \mathbb{R} \rightarrow \mathbb{R}$ given by $g(x) = \begin{cases} x \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Example 7: Thomae's Function: $t : \mathbb{R} \rightarrow \mathbb{R}$ given by $t(x) = \begin{cases} 1 & \text{if } x = 0 \\ \frac{1}{n} & \text{if } x = \frac{m}{n} \in \mathbb{Q} \text{ (in lowest terms, } n > 0) \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$

D_f must be a _____ union of _____ sets. (final project option)

After Class:

Pick & Prove Problems:

★★ Exercise 4.2.1 (Modified)

This exercise aims to illustrate how the Algebraic Limit Theorem for Functional Limits and the Algebraic Continuity Theorem both follow from the Algebraic Limit Theorem for sequences.

- (a) Prove Corollary 4.2.4 part (ii) using the “Sequential Criterion for Functional Limits” and the Algebraic Limit Theorem for sequences.
- (b) Use this result to “prove” part (ii) of Theorem 4.3.4 (it will be very quick).
- (c) Do the same for part (iii).

★★ Supplemental Exercise

Prove that the following functions are continuous on their domains without using the Algebraic Continuity Theorem (use $\epsilon - \delta$ definition.)

- (a) $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 3x + 1$
- (b) $g : \mathbb{R} \rightarrow \mathbb{R}$ given by $g(x) = x^3$

★★ Exercise 4.6.2

Given a countable set $A = \{a_1, a_2, a_3, \dots\}$, define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(a_n) = 1/n$ and $f(x) = 0$ for all $x \notin A$. Find the set of discontinuities of f .

Note: $f(a_n) = 1/n$, not $f(a_n) = 1/a_n$.

★ ★ ★★ Exercises 4.6.5, 4.6.6 (Don’t get too intimidated by the 4 stars, this is more reasonably a 3 star but I didn’t give you any one star problems.)

- (a) Prove that the only type of discontinuity a monotone function can have is a jump discontinuity.
Hint: “This argument is very similar in spirit to the proof of the Monotone Convergence Theorem.”
- (b) Construct a bijection between the set of jump discontinuities of a monotone function and a subset of $\mathbb{Q}$. Conclude that the set of discontinuities of a monotone function must either be finite or countable, but not uncountable.

Understanding Analysis Sections Covered: Section 4.4

Definitions and Theorems to include in Analysis Journal:

- Theorem 4.4.1 (preservation of compact sets)
 - Theorem 4.4.2 (Extreme Value Theorem)
-

Image and Preimage

Given $f : A \rightarrow \mathbb{R}$, we can define the following:

- The **image (range)** of $B \subseteq A$ is $f(B) =$
- The **preimage** of $B \subseteq \mathbb{R}$ is $f^{-1}(B) =$

From homework Exercise 4.3.9: $f^{-1}(\{0\})$ is

Theorem (Preservation of Compact Sets)

Let $f : A \rightarrow \mathbb{R}$ be continuous on A . If $K \subseteq A$ is compact, then...

Proof.

□

Extreme Value Theorem

If $f : K \rightarrow \mathbb{R}$ is continuous and K is compact, then...

Proof.

□

After Class:

Pick & Prove Problems:

★ Exercise 4.4.4ac

Decide whether each of the following statements is true or false, justifying your conclusion.

- (a) If f is continuous on $[a, b]$ with $f(x) > 0$ for all $a \leq x \leq b$, then $1/f$ is bounded on $[a, b]$ (meaning $1/f$ has a bounded range).
- (c) If f is defined on $\mathbb{R}$ and $f(K)$ is compact whenever K is compact, then f is continuous on $\mathbb{R}$.

★ Exercise 4.4.8ab

Give an example of each of the following, or provide a short argument for why the request is impossible. (feel free to draw pictures for examples)

- (a) A continuous function defined on $[0, 1]$ with range $(0, 1)$.
- (b) A continuous function defined on $(0, 1)$ with range $[0, 1]$.

★ ★ ★ Exercises 4.4.11-12 (Topological Characterization of Continuity)

Note: If you've enjoyed the topology, I highly recommend this one.

- (a) Let g be defined on all of $\mathbb{R}$. If B is a subset of $\mathbb{R}$, define $g^{-1}(B)$ by

$$g^{-1}(B) = \{x \in \mathbb{R} : g(x) \in B\}.$$

Show that g is continuous if and only if $g^{-1}(O)$ is open whenever $O \subseteq \mathbb{R}$ is an open set.

- (b) Determine which of the following statements are true about a continuous function defined on $\mathbb{R}$.
 - (i) $f^{-1}(B)$ is finite whenever B is finite.
 - (ii) $f^{-1}(K)$ is compact whenever K is compact.
 - (iii) $f^{-1}(A)$ is bounded whenever A is bounded.
 - (iv) $f^{-1}(F)$ is closed whenever F is closed.

Understanding Analysis Sections Covered: Section 4.5

Definitions and Theorems to include in Analysis Journal:

- Theorem 4.5.1 (Intermediate Value Theorem)
- Theorem 4.5.2 (preservation of connected sets)
- Definition 4.5.3 (Intermediate Value Property)

Before Class:

Readings: None

Analysis Journal Prompts:

- We now know compact sets are preserved under continuous functions. If we now talk about a closed interval (a special case of a compact set), what are its possible images under a continuous function? Could it be any compact set?

Theorem (Preservation of Connected Sets)

Let $f : A \rightarrow \mathbb{R}$ be continuous on A . If $E \subseteq A$ is connected, then...

Scratch Work:

Proof.

□

Intermediate Value Theorem

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. If L is a real number satisfying $f(a) < L < f(b)$ or $f(a) > L > f(b)$, then...

Scratch Work:

Proof.

□

Does the Intermediate Value Theorem have a converse?

Intermediate Value Property

A function f has the intermediate value property on a closed interval $[a, b]$ if for all $x < y$ in $[a, b]$ and all L between $f(x)$ and $f(y)$, then...

If a function has the intermediate value property on a closed interval, is it continuous on that interval?

After Class:***Pick & Prove Problems:*****★ Exercise 4.5.7**

Let f be a continuous function on the closed interval $[0, 1]$ with range also contained in $[0, 1]$. Prove that f must have a fixed point, that is, show $f(x) = x$ for at least one value of $x \in [0, 1]$.

Hint: This is a quick corollary of the intermediate value theorem.

★★ Exercise 4.5.5b

In class we proved the Intermediate Value Theorem as a corollary to the Preservation of Connected Sets. Provide an alternate proof using the Nested Interval Property by following and completing the proof in the book (don't forget to generalize your result to all L at the end, not just $L=0$!)

★★ Exercise 4.5.3

Show that if f is increasing on $[a, b]$ and satisfies the Intermediate Value Property, then f is continuous on $[a, b]$.

Before Class:***Readings:*** None***Analysis Journal Prompts:***

- Finish your Definition and Theorem map for these first two chapters of the textbook. Which proofs rely on which theorems/definitions?
- Come to class with an updated “Questions about Analysis” list. We’ll try to answer some in review for the test.

